

Train Reservation System in C++

```
#include <iostream>
#include <string>
using namespace std;

class Passenger {
public:
    int seatNumber;
    string name;
    Passenger* next;

    Passenger(int seat, string passengerName) {
        seatNumber = seat;
        name = passengerName;
        next = NULL;
    }
};

class TrainReservationSystem {
private:
    Passenger* head;

public:
    TrainReservationSystem() {
        head = NULL;
    }

    void bookTicket(int seat, string name) {
        Passenger* newPassenger = new Passenger(seat, name);

        if(head == NULL) {
            head = newPassenger;
        }
        else {
            Passenger* temp = head;

            while(temp->next != NULL) {
                temp = temp->next;
            }

            temp->next = newPassenger;
        }

        cout << "Ticket Booked Successfully!\n";
    }

    void cancelTicket(int seat) {
        if(head == NULL) {
            cout << "No Reservations Found!\n";
            return;
        }

        Passenger* temp = head;
        Passenger* prev = NULL;

        while(temp != NULL && temp->seatNumber != seat) {
            prev = temp;
            temp = temp->next;
        }

        if(temp == NULL) {
            cout << "Seat Not Found!\n";
            return;
        }

        if(prev == NULL) {
            head = head->next;
        }
    }
};
```

```

        else {
            prev->next = temp->next;
        }

        delete temp;

        cout << "Ticket Cancelled Successfully!\n";
    }

void searchPassenger(int seat) {
    Passenger* temp = head;

    while(temp != NULL) {
        if(temp->seatNumber == seat) {
            cout << "Passenger Found!\n";
            cout << "Seat Number: " << temp->seatNumber << endl;
            cout << "Passenger Name: " << temp->name << endl;
            return;
        }

        temp = temp->next;
    }

    cout << "Passenger Not Found!\n";
}

void displayReservations() {
    if(head == NULL) {
        cout << "No Reservations Available!\n";
        return;
    }

    Passenger* temp = head;

    cout << "\n===== RESERVED TICKETS =====\n";

    while(temp != NULL) {
        cout << "Seat Number: " << temp->seatNumber
            << " | Passenger Name: " << temp->name << endl;

        temp = temp->next;
    }
}

void menu() {
    int choice;
    int seat;
    string name;

    do {
        cout << "\n===== TRAIN RESERVATION SYSTEM =====\n";
        cout << "1. Book Ticket\n";
        cout << "2. Cancel Ticket\n";
        cout << "3. Search Passenger\n";
        cout << "4. Display Reservations\n";
        cout << "5. Exit\n";

        cout << "Enter Your Choice: ";
        cin >> choice;

        switch(choice) {
            case 1:
                cout << "Enter Seat Number: ";
                cin >> seat;

                cin.ignore();

                cout << "Enter Passenger Name: ";
                getline(cin, name);

                bookTicket(seat, name);

```

```

        break;

    case 2:
        cout << "Enter Seat Number to Cancel: ";
        cin >> seat;

        cancelTicket(seat);
        break;

    case 3:
        cout << "Enter Seat Number to Search: ";
        cin >> seat;

        searchPassenger(seat);
        break;

    case 4:
        displayReservations();
        break;

    case 5:
        cout << "Exiting System...\n";
        break;

    default:
        cout << "Invalid Choice!\n";
    }
} while(choice != 5);
};

int main() {
    TrainReservationSystem system;

    system.menu();

    return 0;
}

```